

SABARI M

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EXECUTIVE SUMMARY

Electronics and Communication Engineering student with hands-on experience in PCB design and exploring analog circuits. Proficient in Altium Designer and STM32 development tools.

SKILLS

PCB Design: Altium Designer, Schematic capture, Circuit Design, PCB layout Design, DRC, Gerber generation, Output job files

Embedded Systems: STM32, ESP32, Sensor calibration

Programming Language: C, MATLAB

Tools & Tech: STM32CubeIDE, Altium Designer, KiCAD, Arduino IDE, STM32CubeMonitor, Keil uVision, VS Code, ANSYS HFSS, Verilator.

INTERNSHIP EXPERIENCE

Punha LLP, PCB Design intern 03/2026 – 06/2026
Remote

- Working in designing circuits and PCB for healthcare wearables
- Explored ADC ICs for designing PCBs for prosthetic wearable devices

ISRO - LIQUID PROPULSION SYSTEM CENTER, Student intern 10/2025 – 11/2025
Tiruvandrum, Kerala

Intern – Flow Measurement & Sensor Calibration

- Worked on calibration of **Coriolis** and **Thermal Mass Flow Meters** sensors using standard **instrumentation** practices.
- Studied flow measurement chains including **sensors, transmitters, signal conditioning, TIB, and DAQ systems**.
- Gained hands-on exposure to **analog signal processing, sensor interfacing, and industrial instrumentation electronics**.

PROJECTS

CUSTOM ESP32S3 MINI WITH USB TYPE C - PCB DESIGN 

- Designed a custom ESP32-S3 Mini development board integrating USB Type-C for efficient power and data transfer.
- Optimized the PCB for compact form factor and manufacturability, targeting IoT, embedded systems, and wireless applications.

Orientation Control Using MPU6050 by STM32F401CCU6 MCU 

- Acquired raw accelerometer and gyroscope data, applied offset calibration, and calculated roll angle from accelerometer readings.
- Implemented a Kalman filter in C to fuse accelerometer and gyroscope data, achieving smooth and precise tilt angle estimation.

Automatic Temperature and Water Controlling System

- Developed a microcontroller-based system to monitor and control temperature and water levels using appropriate sensors.
- Simulated and validated the system design using TinkerCAD before hardware implementation.

CERTIFICATES

- Altium Skilled Engineer 
- Altium Designer Essential - On demand 
- PCB Design Essentials - Altium 
- Embedded Systems 
- Altium Emerging Engineer 
- Altium Workspace User Management 
- C Programming Hands On : Skill Rack 

EDUCATION

Rajalakshmi Institute of Technology, Chennai, 2023 – 2027
B.E , Electronics and Communication Engineering

Sir .C. P. Memorial Government Higher Secondary School 2016 – 2023
Kanyakumari, VI to XII , HSC Percentage -89%

VOLUNTEERING

Rajalakshmi Institute of Technology, IEEE Student Branch Treasurer 09/2025 – Present

IEEE Antennas and Propagation Society, Student Member 06/2025 – Present